# EE/CE 252/251
# Signals and Systems

with

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@

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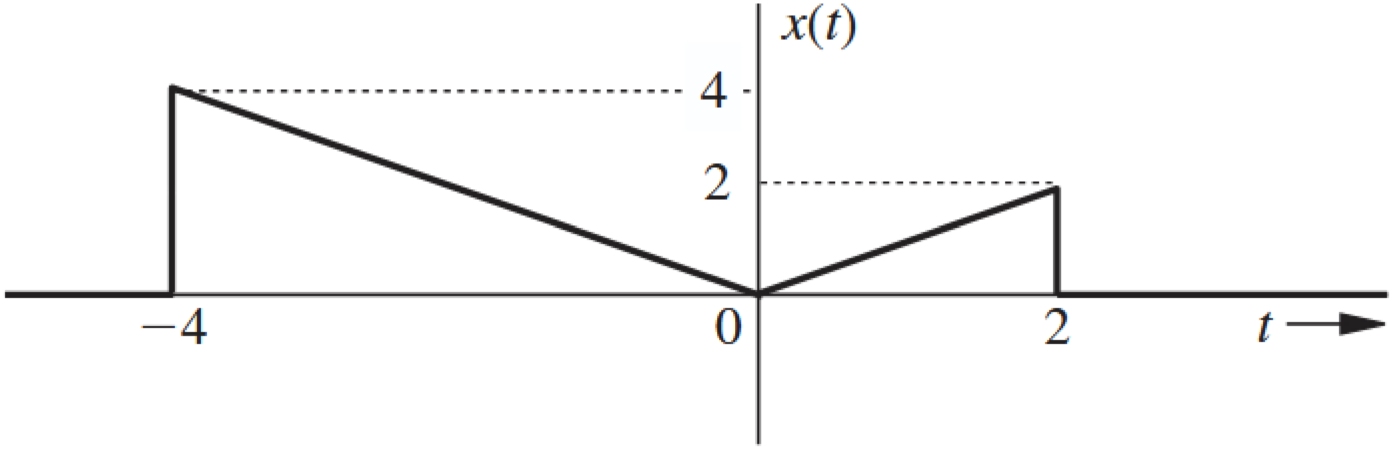
## Problem 1

For the signal shown plot the following

$x(-t)$

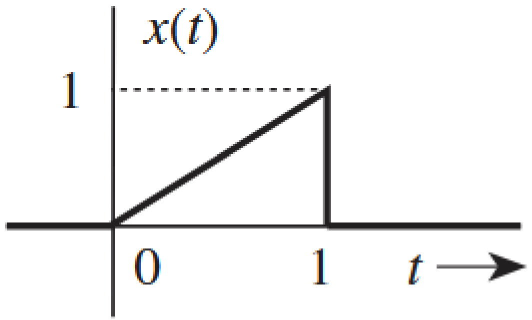
$x(2t-4)$

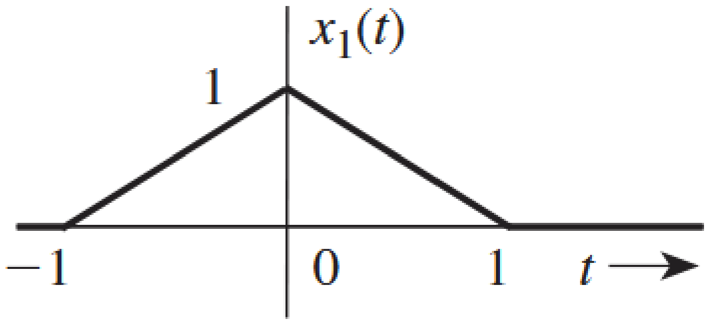
$x(2-t)$

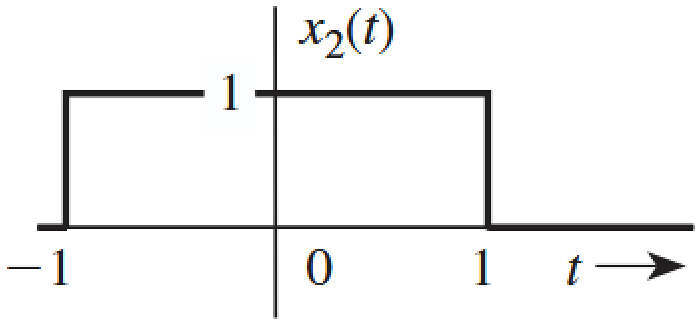


## Problem 2

Write $x_1, x_2$ in terms of $x$



## Problem 3

Consider an energy signal $x(t)$. Which of the following operations will not change its energy?

1. Time-shifting
2. Time-reversal
3. Time-compression
4. Time-expansion
5. Amplitude-scaling

Justify your claims.

# Problem 4

If $x_e(t)$ and $x_o(t)$ are even and the odd components of a real signal $x(t)$, then show that

$$\int_{-\infty}^{\infty} x_e(t)x_o(t)\,dt = 0$$

## Problem 5

Find and sketch the odd and the even components of the following:

(b) $tu(t)$

(c) $\sin \omega_0 t$

# Problem 6

Given $x_1(t) = \cos(t)$, $x_2(t) = \sin(\pi t)$, and $x_3(t) = x_1(t) + x_2(t)$.

Is $x_3(t)$ periodic?

## Problem 7

Simplify the following expressions:

(a) $\left(\dfrac{\sin t}{t^2+2}\right)\delta(t)$

(b) $\left(\dfrac{j\omega+2}{\omega^2+9}\right)\delta(\omega)$

(c) $[e^{-t}\cos(3t-60°)]\delta(t)$

(d) $\left(\dfrac{\sin\left[\frac{\pi}{2}(t-2)\right]}{t^2+4}\right)\delta(1-t)$

(e) $\left(\dfrac{1}{j\omega+2}\right)\delta(\omega+3)$

(f) $\left(\dfrac{\sin k\omega}{\omega}\right)\delta(\omega)$

## Problem 8

Evaluate the following integrals:

(a) $\int_{-\infty}^{\infty} \delta(\tau)x(t-\tau)\,d\tau$

(b) $\int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)\,d\tau$

(c) $\int_{-\infty}^{\infty} \delta(t)e^{-j\omega t}\,dt$

(d) $\int_{-\infty}^{\infty} \delta(2t-3)\sin \pi t\,dt$

(e) $\int_{-\infty}^{\infty} \delta(t+3)e^{-t}\,dt$

(f) $\int_{-\infty}^{\infty} (t^3+4)\delta(1-t)\,dt$

(g) $\int_{-\infty}^{\infty} x(2-t)\delta(3-t)\,dt$

## Problem 9

Find area under the given functions for $-2 < t < 3$

$$\delta(2t-3)\sin \pi t\, dt$$

$$\delta(t+3)e^{-t}\, dt$$

$$(t^3+4)\delta(1-t)\, dt$$

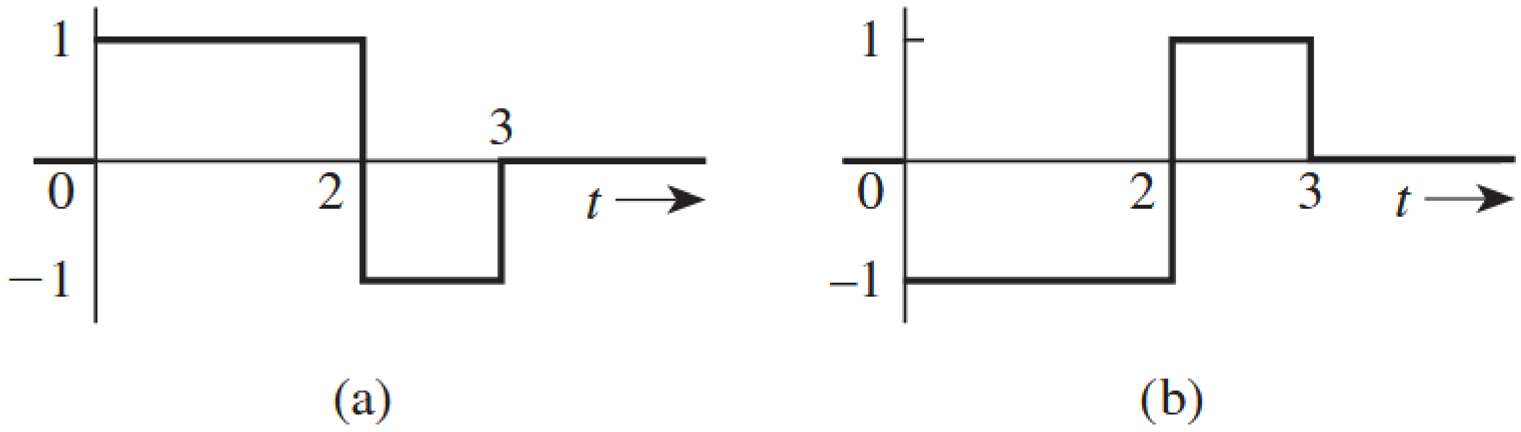
**Problem 10**

$$x(t) = De^{j\omega_0 t}$$

Is this a power or energy signal? Find both.

## Problem 11

Are the following signals energy signals or power signals? Find their energy and power.



# Questions?? Thoughts??